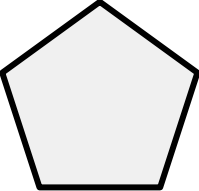


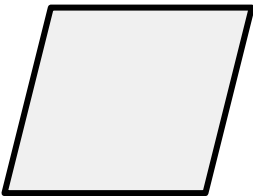


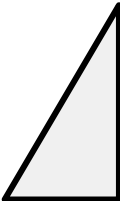
Rotational symmetry

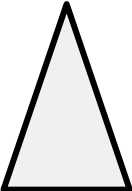
Task A: Identifying rotational symmetry in shapes

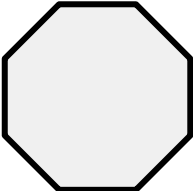
1) Write down the order of **rotational symmetry** for each shape. Tracing paper may help you with this task.

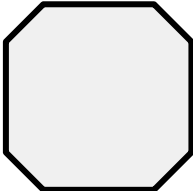
a)

5

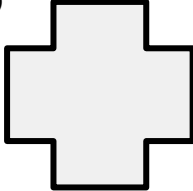
b)

2

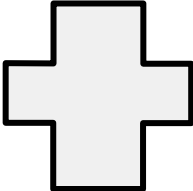
c)

1

d)

1

e)

8

f)

4

g)

4

h)

2

2) Sort these letters into the table, according to their order of **rotational symmetry**.

A B C D E F G H I J K L M
N O P Q R S T U V W X Y Z

order 1	order 2	order 3	order 4	order greater than 4
A, B, C, D, E, F, G, J, K, L, M, P, Q, R, T, U, V, W, Y	H, I, N, S, X, Z			O

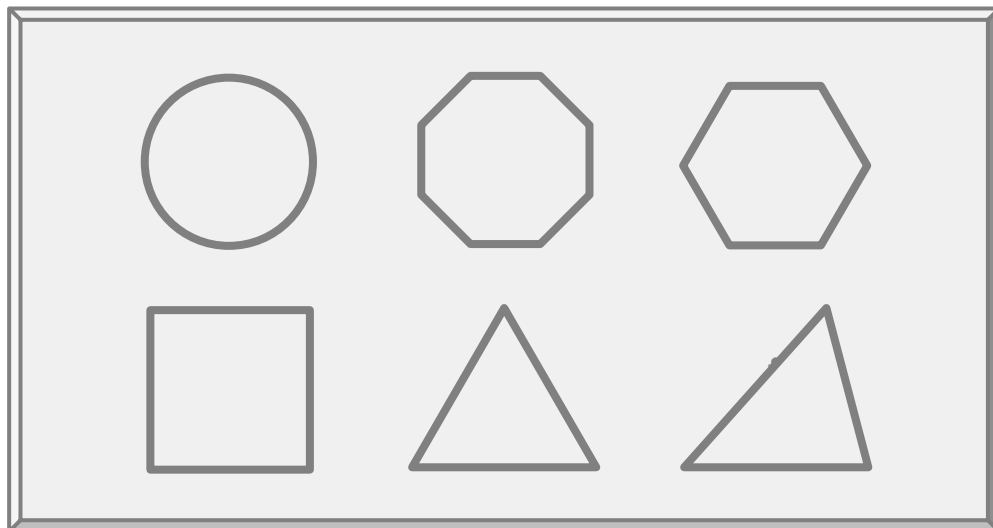
Note: rotational symmetry for letters O and X may differ depending on font.

Task B: Finding shapes with particular rotational symmetries

- 1) Aisha is designing a puzzle board for preschool children, where they place shapes into spaces on the board. She wants to arrange the spaces so that the shapes are in descending order of **rotational symmetry**.

Example:

- a) circle
- b) regular octagon
- c) regular hexagon
- d) square
- e) equilateral triangle
- f) scalene triangle

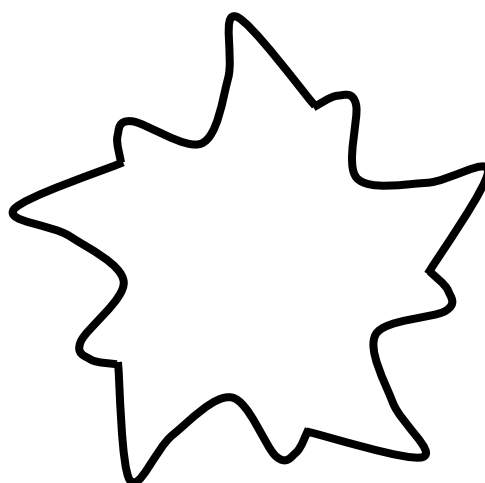
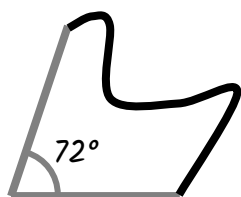


Draw six shapes onto the board so that each shape has a lower order of **rotational symmetry** than the previous one.

- 2) Design and draw a custom logo with **rotational symmetry** of order 5. Tracing paper and a protractor may be helpful during this task. Alternatively, you may want to use computer software.

Vertex at the centre would have an angle of size 72° .

Example:



Answers can be checked using tracing paper.

